Model Predictive Control
Prof. M. Zeilinger

Spring 2026

Exercise 11
Iterative Optimization

Exercise 1

Consider the nonconvex optimization problem

$$\min_x -e^{-x^2} \tag{1}$$

1. Starting from $x_0 = -0.1$, derive the first two iterations of the gradient method with h = 1.
2. Starting from $x_0 = -0.1$, derive the first two iterations of the Newton method with h = 1.
3. Implement the gradient method and Newton's method.
4. Determine *numerically* the region of convergence of Newton's method and show for which value of x_0, the Newton direction (with the appropriate step size) is a descent direction.
5. Determine *numerically* the region of convergence of the gradient method and show for which value of x_0, gradient descent (with the appropriate step size) leads to descent directions.
6. Show *numerically* that Newton's method converges faster than the gradient method.

Exercise 2

Consider the following optimization problem

$$\begin{aligned} \min_{x\in\mathbb{R}^2} \quad & \frac{1}{2}(x-v)^T Q(x-v) \\ \text{s.t.} \quad & ||x||_2 \leq 1 \end{aligned} \tag{2}$$

Find numerically the optimal solution using the projected method with $v = [2, 3]^T$ and $Q = \text{diag}(1, 2)$.